

RUTUJA PATIL

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Education

B.E. Computer Engineering
Saraswati College of Engineering, Navi Mumbai

2022 – 2026
CGPA: 8.91

Experience

Intern — Full Stack Developer & ML Engineer
Vortex Web Technologies, Nerul

Jan – Apr 2026

- Engineered and integrated a MySQL backend for a healthcare consultation platform using RESTful APIs and Node.js for seamless frontend-backend communication.
- Developed a full-stack agriculture marketplace platform enabling farmers, buyers, and transporters to manage product workflows and KYC verification.
- Integrated a plant disease detection ML model using FastAPI and Node.js for automated crop analysis and intelligent prediction capabilities.

Projects

FakeXpose — Multimodal Deepfake Detection System

Python, TensorFlow, FastAPI, OpenCV

- Developed a multimodal deepfake detection system using EfficientNet and ensemble learning models for robust video and audio classification.
- Implemented frame-level face extraction and weighted aggregation using OpenCV and MediaPipe, achieving up to 95% video detection accuracy.
- Built an audio analysis pipeline with 170-dimensional feature extraction and ensemble classifiers, achieving 97.5% classification accuracy.

SIGRO — AI-Powered Smart Agriculture Assistant

React.js, Node.js, Python, YOLOv8

- Trained and deployed a YOLOv8-based plant disease detection model with ONNX optimization for real-time crop analysis and prediction.
- Designed a Deep Reinforcement Learning engine analyzing crop, soil, and weather conditions to optimize irrigation and fertilizer strategies.
- Developed a multilingual AI dashboard integrating voice assistance, analytics, and real-time monitoring for agricultural management.

MONEY-MITRA — AI Financial Management Platform

MERN Stack, JWT, Chart.js

- Built a secure AI-driven financial management platform for expense tracking, budgeting, and savings management through an interactive dashboard.
- Implemented JWT authentication, RBAC, and intelligent financial scoring modules to improve platform security and personalization.
- Developed analytics and visualization modules using Chart.js for AI-generated spending insights and predictive budgeting recommendations.

Technical Skills

Languages:	Java, JavaScript, Python
Web Technologies:	MERN Stack, HTML, CSS, Tailwind CSS, React.js
Tools & Platforms:	Git, GitHub, Docker, Vercel, Vite
AI/ML:	Deep Learning, LLMs, Chatbots

Achievements

Hackathons: Secured 1st Prize in Software Category at Avishkar 2026.
Secured 2nd Runner-Up at Loop Hackathon among 2500+ participating teams.

Events: Participated in CodeCrafters 2.0, Mumbai-Hacks, Quasor 2.0, and Hack4 Innovations.

Training: CNC Web World — Full Stack Development Training.